Binsiya T.N

Embedded Software Engineer | IIoT & Firmware Developer | STM32, ESP32, Arduino | Industrial Automation & Energy Systems

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Professional Summary

Embedded Software Engineer with 2+ years of experience building firmware for STM32, ESP32 and Arduino-based products and IIoT gateways, taking solutions from concept to commissioning.

- M.Tech in Applied Electronics (CGPA 8.7) and PG Diploma in Embedded Systems.
- Firmware on STM32 (STM32CubeIDE, HAL/LL) and ESP32 (ESP-IDF, Arduino framework) with UART, SPI, I2C, CAN/J1939, RS-485 (Modbus RTU), Modbus TCP, DLMS and MQTT.
- Delivered Hybrid Controller Pack for power systems (diesel generators, batteries, BMS and rectifiers) at telecom towers in Dubai.
- Implemented OTA firmware updates and device management, which cut manual site visits and downtime.
- Scripting expert: built customizable control logic with Wrench Script, so site-specific behavior changes without a firmware update.
- Works in Agile/Scrum with Git and Azure DevOps; reads schematics and datasheets and works closely with hardware teams on board bring-up and validation.

Technical Skills

- **Languages:** C, Embedded C, C++, Script (customizable logic without firmware update)
- **Microcontrollers & IDEs:** STM32 / ARM Cortex-M (STM32CubeIDE, HAL/LL), ESP32 (ESP-IDF, Arduino framework), Arduino (Nano, rapid prototyping), PIC16F877A (MPLAB)
- **Communication Protocols:** UART, SPI, I2C, CAN Bus (CAN/J1939), RS-485 (Modbus RTU), Modbus TCP, DLMS, MQTT, HTTP, Ethernet (basic)
- **Peripherals & Memory:** ADC (analog sensor acquisition), EEPROM, Flash memory, LCD/HMI displays, GSM, GPS, LTE, I/O extenders
- **Firmware Engineering:** Bootloader, OTA firmware updates, JSON, device management, debugging, code and memory optimization
- **RTOS & OS:** FreeRTOS, Embedded Linux (basic)
- **Debugging Tools:** logic analyzer, oscilloscope, CAN tools
- **IIoT & Platforms:** MicrEdge gateway, ThingsBoard, AWS
- **Sensors & Actuators:** Analog, digital, MEMS sensors, stepper, servo and brushed DC motors (basic)
- **Hardware:** Schematic and datasheet reading, board bring-up and validation, electronics troubleshooting
- **Testing & VCS:** Git, Git Bash, GitHub, Azure DevOps, Agile/Scrum, unit, integration and system testing, peer reviews, release management
- **Domain:** BMS, industrial automation, IoT products, energy systems

Experience

Self-Directed R&D | Embedded Systems Engineer — September 2025 to Present

- Researching firmware performance optimization and emerging IIoT technologies.
- Developing firmware on STM32 (STM32CubeIDE, HAL/LL) and ESP32 (ESP-IDF) for hardware-software integration challenges.

Consyst Digital Industries | Embedded Software Engineer — April 2024 to April 2025, Kochi, India

- Developed firmware for customized IIoT gateways monitoring hybrid energy systems (diesel generators and deep-cycle batteries) in real time.
- Implemented OTA firmware update and device management, reducing manual site visits and system downtime.
- Built customizable control logic with Wrench Script, allowing logic changes to be deployed in the field without a firmware update.
- Integrated CAN/J1939, RS-485 (Modbus RTU), DLMS and MQTT for device interoperability.
- Ran unit, integration and system-level testing with simulation and debugging tools.
- Managed the release pipeline in Agile/Scrum using Git and Azure DevOps, supported R&D and project commissioning.
- Wrote design and test documentation for compliance and future scalability.

Consyst Digital Industries | Intern, Embedded Software Engineer — January 2024 to March 2024, India

- Carried out board bring-up and hardware validation, created test plans and design documentation.
- Released software in line with quality standards and debugged issues with cross-functional teams.

Nest Cyber Solutions | Trainee, Embedded Systems — October 2022 to November 2023, India

- Integrated microcontrollers with industrial sensors and actuators.
- Optimized code for memory use, power efficiency and performance.
- Debugged and resolved software issues in a multidisciplinary team.

Projects

Hybrid Controller for Telecommunication Systems (demonstrated at TRL 6)

- Designed, developed and commissioned Hybrid Controller Pack across telecom towers in Dubai.
- Integrated BMS and rectifier modules over RS-485 (Modbus RTU) and controlled diesel generator operation over CAN J1939.
- Connected hardwired I/Os, HMI displays and alarm patching, sent data to the IIoTNext platform over LTE and wireless modules.
- Created Wrench Script for customizable logic, updated without reflashing firmware.
- Stack: ESP32, Embedded C/C++, Wrench Script, Modbus RTU, CAN J1939.

Solar Plants Monitoring System over the Internet (KSEB, Kerala)

- Designed and developed the MicrEdge IIoT gateway for remote monitoring and control of solar plants.
- Acquired energy meter data over DLMS and Modbus, integrated wireless, LTE and GSM connectivity with real-time configuration.
- Enabled remote gateway updates through device management and project management.
- Stack: ESP32, Embedded C/C++, DLMS, Modbus, GSM, LTE.

LPG Tank Level and Pressure Monitoring System

- Built an IIoT gateway that reads sensor data and sends it to the ThingsBoard platform.
- Stack: MicrEdge gateway, ThingsBoard.

Education

- PG Diploma, Embedded Systems — Nest Cyber Campus, 2022 to 2023
- Master of Technology, Applied Electronics — APJ Abdul Kalam Technological University, 2020 (CGPA 8.7)
- Bachelor of Technology, Electronics and Communication Engineering — M.G. University, 2018 (CGPA 7.9)

Awards

- Appreciation letter for the Hybrid Controller project in the U.A.E.

Certifications

- IELTS (B2 TRF), 2022
- Networking certificate, Cisco Networking Academy, September 2026

Languages: English, Hindi, Tamil, Arabic (limited working)